Roll No.							

End Semester Examination 2024

Name of the Course: Diploma (CSE) — *Semester:* VI

Name of the Paper: Information Security — *Paper Code:* DTCS-603

Time: 3 Hour's — Maximum Marks: 100

Note:

(i) All Questions are compulsory.
(ii) Answer any two sub questions among a, b, and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(10 X 2 = 20 Marks)	**CO1**
(a)	Discuss the following terms: (2 X 5 = 10 marks) (i) DOS, DDOS and sniffing (ii) Backdoors, trapdoors, and spoofing	
(b)	List various threats to security. Explain encryption attacks in detail explaining indirect, dictionary, brute force and birthday attacks.	
(c)	Explain the following terms: (2 X 5 = 10 marks) (i) Intruders and insiders (ii) Viruses and its types	
Q2	(10 X 2 = 20 Marks)	**CO2**
(a)	State computer security challenges. Explain Classic security models such as Bell- La padula, Biba and Clarke Wilson model using appropriate diagrams and examples.	
(b)	Clarify the following terms: (2 X 5 = 10 marks) (i) Role of people in security: Guest user and cyber-security professional (ii) Layered security and multiple layers used to provide security	
(c)	What is Access control? Discuss how access control works and what are the benefits of using access control methods? Explain Role based access control method.	
Q3	(10 X 2 = 20 Marks)	**CO3**
(a)	Answer the following terms in detail: (2 X 5 = 10 marks) (i) Dumpster diving (ii) Physical access by non-employees	
(b)	What is Piggybacking? Describe how an unauthorized person can use piggybacking to get an access. How you can solve this problem using man trap? Also discuss security awareness people should follow to avoid these situations.	
(c)	State the following terms: (2 X 5 = 10 marks) (i) Installing unauthorized software and hardware (ii) Password selection	
Q4	(10 X 2 = 20 Marks)	**CO4**
(a)	What are security policies, procedures, standards and guidelines? Describe policy life cycle mentioning proper steps.	

(b)	Explain the following terms: (2 X 5 = 10 marks) (i) Physical Security (ii) Social Engineering	CO4
(c)	What is Biometrics? Explain working of biometric system using block diagram of Biometric system. List basic types of biometrics. Discuss Facial biometrics and Voice biometrics in detail.	
Q5	(10 X 2 = 20 Marks)	CO5
(a)	What is Cryptography? Explain characteristics of cryptographic systems. Also discuss symmetric cipher model.	
(b)	Describe the working principle of Caesar's cipher and illustrate it with an example. How does it differ from a simple shift cipher? Debate its vulnerability to brute-force attacks and its limitations in modern cryptographic contexts.	
(c)	State the Data Encryption Standard (DES) and its significance in symmetric encryption with the help of its structure. Discuss the vulnerabilities of DES, leading to the development of Advanced Encryption Standard (AES).	